Student handbook

Every lesson in one booklet

This handbook follows the online course lesson by lesson. Use it in class, at home, or offline. The videos, labs and quizzes are online.

Course map

Module	Lessons
What is AI? Where AI is around you, how learning from examples differs from following rules, and the data every AI learns from.	1. Welcome: AI is all around you (12 min) 2. Rules or learning? How AI learns (14 min) 3. Data: features and labels (14 min)
Train your own AI How computers see pictures, training your own image classifier, how it decides, and testing whether it is any good.	4. How computers see: pixels (13 min) 5. Train your first AI model (18 min) 6. How your AI decides: nearest neighbours (14 min) 7. Testing: is your AI any good? (18 min)
AI that reads and talks Decision trees, teaching AI to read feelings in words, how chatbots predict the next word, and talking to a real AI safely.	8. Decision trees: AI that asks questions (15 min) 9. Teaching AI to read: words and feelings (15 min) 10. How chatbots write: predicting the next word (15 min) 11. Talk to a real AI, safely (18 min)
Fair, safe and yours How unfair data makes unfair AI, using AI responsibly, and building and presenting your own AI project.	12. Fair AI: when data is unfair (16 min) 13. Responsible AI: privacy, deepfakes and you (15 min) 14. Final project: build and present your AI (30 min) 15. Final assessment and celebration (15 min)

Welcome: AI is all around you

In this course you will build your own AI: one that learns to recognise drawings and objects you show it, right here in your browser. Along the way you will find out how AI really works, where it goes wrong, and how to use it safely and fairly.

You will:

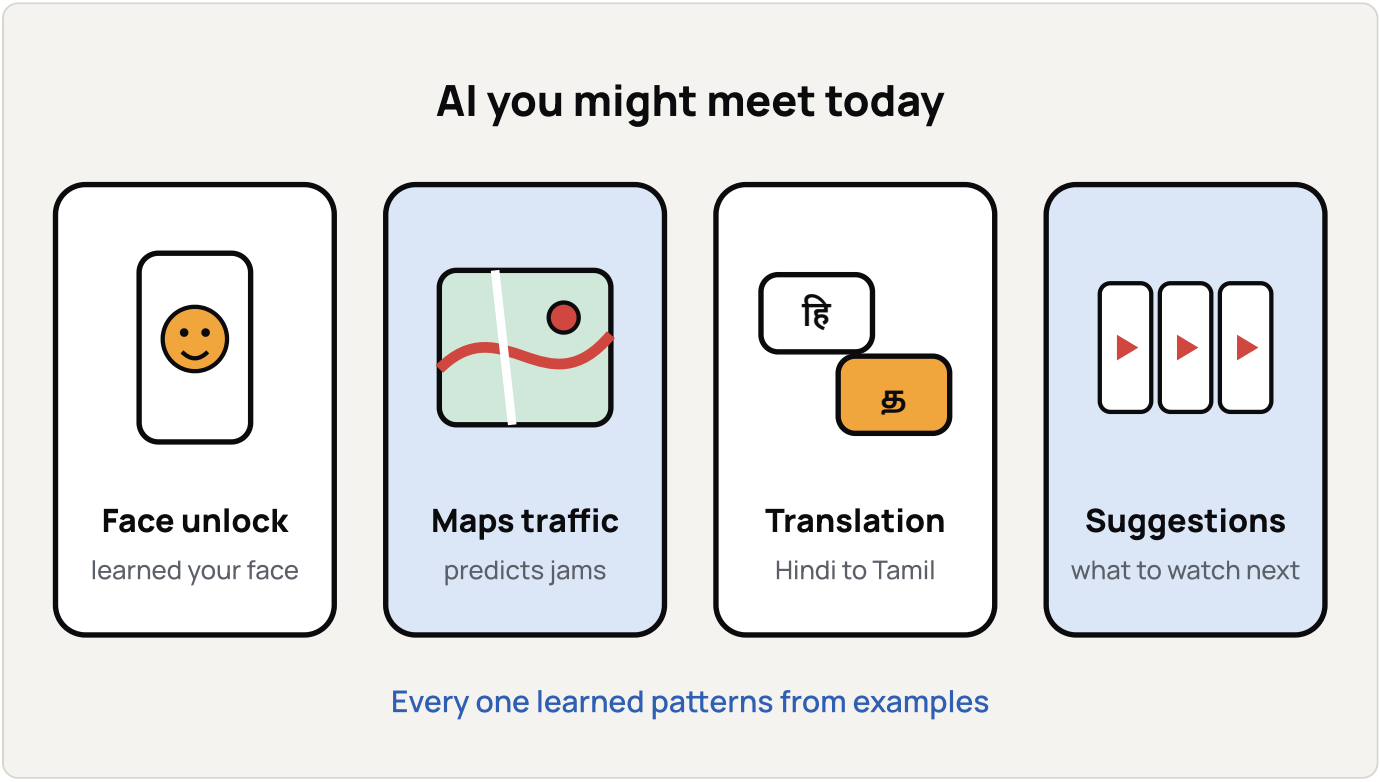
- Say what artificial intelligence (AI) is in your own words
- Spot AI in everyday life in India
- Know the five Safe AI rules for this course
- Know what you will build by the end

What is AI?

Artificial intelligence (AI) is when a computer does something we usually think needs a human brain: recognising a face, understanding speech, translating a language or suggesting the next video.

Most AI today works the same way: instead of a person writing every rule, the computer **learns from lots of examples**. Show it thousands of photos labelled "cat" and "dog", and it learns the patterns that tell them apart.

AI is not magic and it is not alive. It is a pattern-finder, built by people, that can be very useful and can also be wrong.



Four places you might have met AI today.

AI in India

Banks use AI to spot suspicious UPI payments in a fraction of a second. **Bhashini**, a Government of India project, uses AI to translate between Indian languages. Farmers photograph a leaf and an app suggests which crop disease it might have. Maps apps predict traffic so you reach school on time.

What you will build

By the end of this course you will have trained your own **image classifier**: an AI that looks at a drawing (or an object you hold up to the camera) and says which group it belongs to. You will collect the examples, train it, test it, find its mistakes and make it fairer.

You will also talk to a real AI chatbot, safely, and find out how it writes its answers one word at a time.

Five Safe AI rules for every lesson

1. **Keep personal details private.** Never type your full name, address, phone number, school name or passwords into an AI.
2. **AI can be wrong.** Check important facts with a book, a teacher or a trusted website.
3. **Be kind.** Never use AI to make fun of, trick or hurt anyone.
4. **Be honest.** Say when you used AI to help with your work.
5. **Ask an adult** if an AI says something that worries or upsets you.

What you need

A computer, tablet or phone with a modern browser. A mouse or touch screen helps for drawing. A webcam is optional: every camera activity can also be done by drawing. Everything you train stays on your device.

What you learned

- AI is a computer doing tasks that usually need human thinking, mostly by learning patterns from examples.
- AI is all around you: face unlock, translation, maps, fraud checks and recommendations.
- AI can be wrong, so keep personal details private and check important facts.
- You will train your own image classifier and talk to a real chatbot safely.

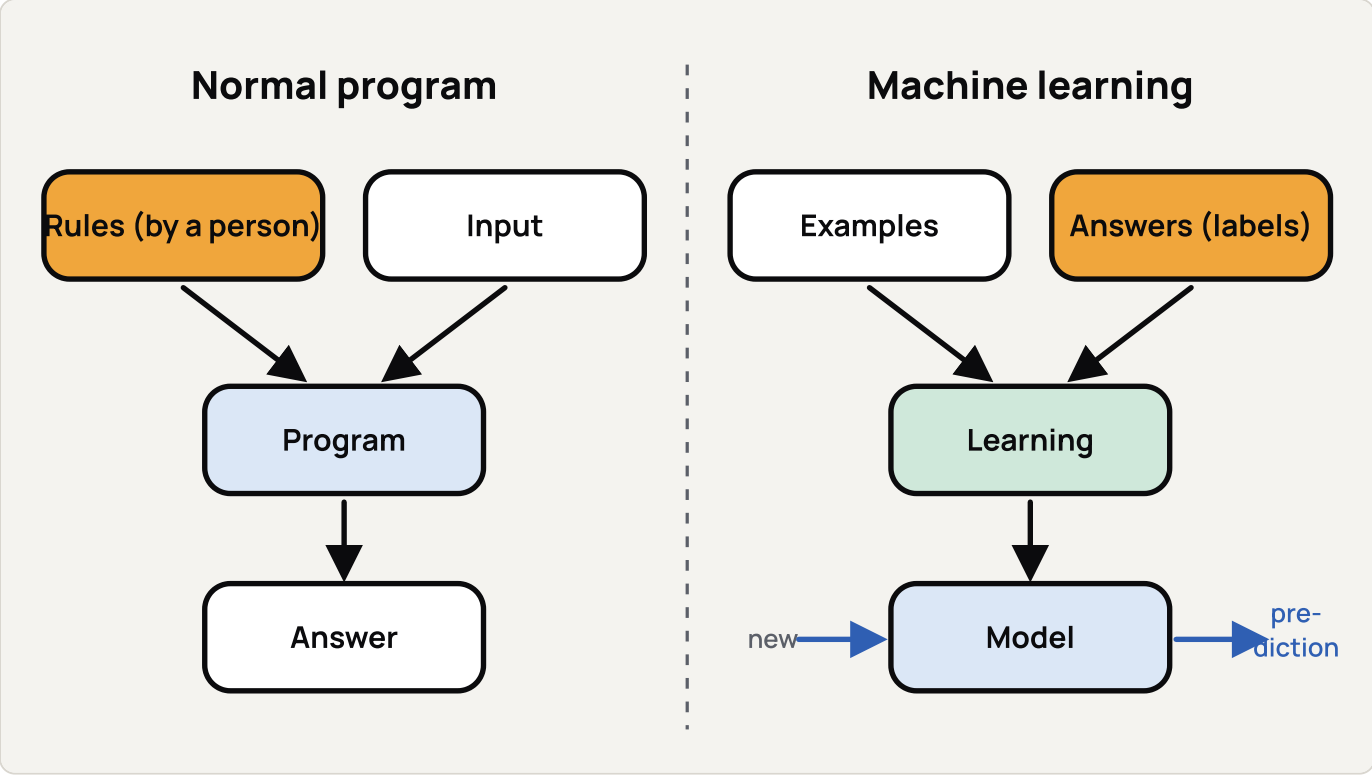
Next: Next, find out the big difference between a computer that follows rules and one that learns.

Rules or learning? How AI learns

Normal programs follow rules a person wrote. Machine learning flips this around: you give the computer examples and the right answers, and it works out the rules for itself. This one idea powers almost every AI you use.

You will:

- Explain the difference between a rules program and machine learning
- Put the five steps of machine learning in order
- Decide when rules are enough and when learning is better



Left: a person writes the rules. Right: the computer learns them from labelled examples.

Two ways to make a computer smart

Rules. A person writes exact instructions: “if the temperature is above 30, switch on the fan.” This works brilliantly when the rules are simple and clear.

Learning. Try writing rules for “is this a photo of a cat?” Cats can be big, small, orange, striped, sleeping or jumping. You would never finish. Instead, you show the computer thousands of labelled photos and let it find the patterns. That is **machine learning**, and the thing it builds is called a **model**.

New words

Machine learning: a computer learning patterns from examples instead of being given every rule.

Model: what the computer builds when it learns. You give it a new example and it gives you a prediction.

Training: the learning step, when the model looks at the examples.

Prediction: the model's answer for something new.

What you learned

- Rules programs follow instructions a person wrote.
- Machine learning finds the rules from labelled examples and builds a model.
- The steps are: collect, label, train, test, use.
- Simple, exact jobs suit rules; messy, varied jobs suit learning.

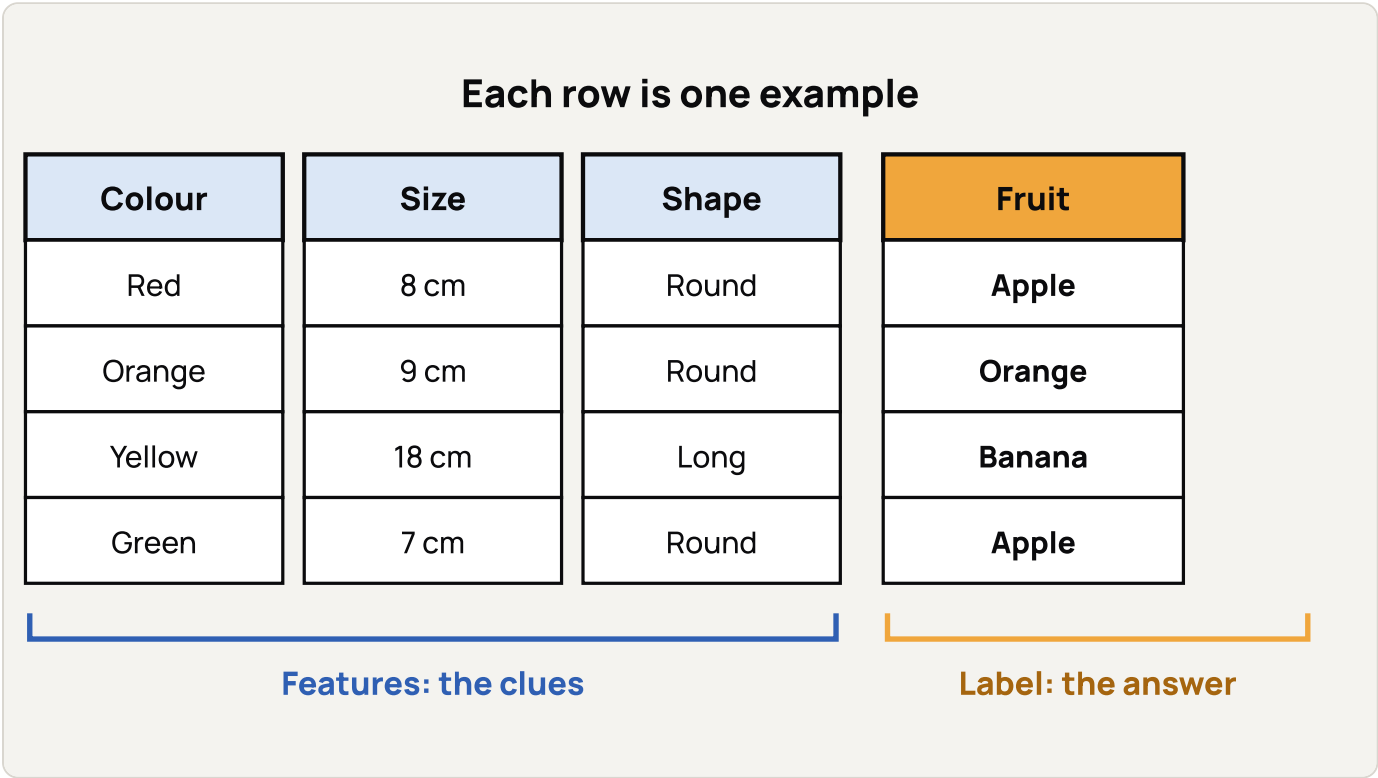
Next: Next, look closely at the examples themselves: the data, its features and its labels.

Data: features and labels

An AI is only as good as the examples it learns from. In this lesson you will learn to describe data the way an AI sees it: as features (the clues) and labels (the answers).

You will:

- Explain what data, features and labels are
- Pick useful features for a task
- Explain why more, and more varied, data helps



Each row is one example. The features are the clues; the label is the answer the AI should learn.

Clues and answers

Data is the collection of examples. Each example has:

- **Features:** measurable clues, like colour, size, weight, or the pixels of a picture.
- **A label:** the right answer, like “apple” or “orange”.

During training, the model looks for how the features connect to the labels. Good features make that easy: colour is a great clue for apples versus oranges. Bad features make it hard: the day of the week the fruit was bought tells you nothing.

More data, and different data

If every “dog” photo shows a brown dog on grass, the AI might learn “brown thing on grass = dog” and fail on a white dog indoors. Good data has **lots of examples** and **lots of variety**: different colours, sizes, angles, lighting and backgrounds.

What you learned

- Data is a set of examples; each has features (clues) and a label (answer).
- Good features connect to the label; useless ones add noise.
- More examples and more variety make an AI work better in real life.

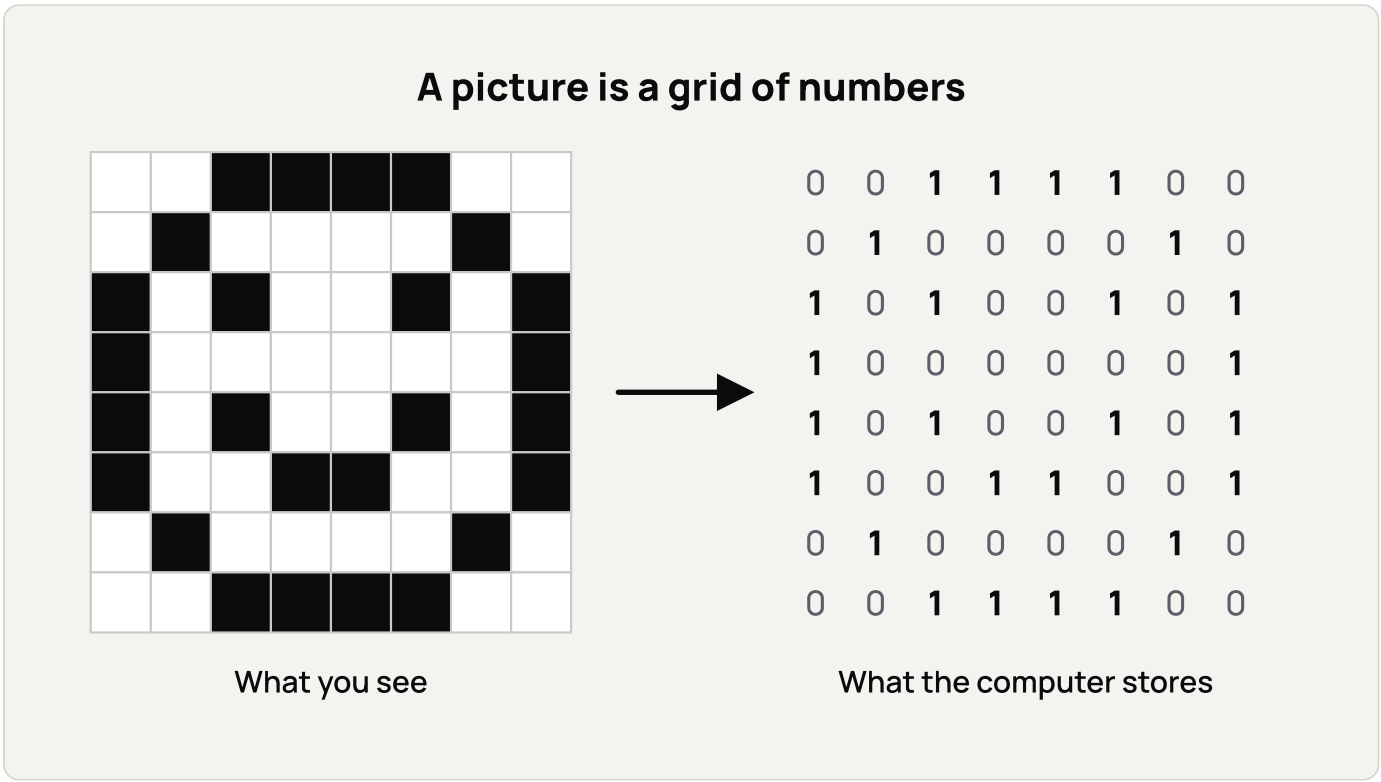
Next: Next, discover how a computer sees a picture: as a grid of numbers called pixels.

How computers see: pixels

You see a smiley face. A computer sees a grid of numbers. Before you can train an AI on pictures, you need to know what a picture looks like from the inside.

You will:

- Explain what a pixel is
- Turn a picture into numbers and numbers back into a picture
- Explain why an AI needs pictures to be numbers



Zoom in far enough and every picture is a grid of squares. Each square is stored as a number.

Pictures are numbers

A digital picture is a grid of tiny squares called **pixels**. Each pixel is stored as a number.

- In a black-and-white picture, 0 can mean white and 1 can mean black.
- In a grey picture, each pixel is a number from 0 (white) to 255 (black).
- In a colour picture, each pixel has three numbers: how much **red**, **green** and **blue** light to mix.

A phone photo has about 12 million pixels, so 36 million numbers! An AI never sees “a cat”. It sees those numbers, and learns which patterns of numbers usually mean “cat”.

▶ Online lab: paint pixels and decode secret pictures from their numbers.

New words

Pixel: one tiny square of a digital picture (short for “picture element”).

Resolution: how many pixels a picture has, such as 1920 × 1080.

RGB: red, green and blue, the three numbers that make each colour pixel.

What you learned

- A digital picture is a grid of pixels, and each pixel is stored as a number.
- Black-and-white pixels can be 0 or 1; colour pixels have red, green and blue numbers.
- An AI learns patterns in those numbers, not “pictures” the way we see them.

Next: Next, use what you know about pixels to train your very first AI model.

Train your first AI model

Time to build! You will teach an AI to recognise simple drawings. You choose the groups, you draw the examples, you press Train, and then you see if it can recognise drawings it has never seen.

You will:

- Collect labelled examples for two or more groups
- Train an image classifier in your browser
- Try your model on new drawings and explain what it gets right and wrong

How to train your model

1. **Choose your groups.** Start with the suggested groups (circle, triangle, square) or rename them. Simple, very different shapes work best for your first model.
2. **Draw examples.** Pick a group, draw one example and press Add example. Draw at least 5 for every group.
3. **Train.** Press Train. Your model learns from every example you drew. This happens on your device; nothing is sent anywhere.
4. **Try it.** Draw something new and see what your model predicts, and how sure it is.

▶ **Online lab: train your own model by drawing or with the camera, then test it.**

Make it smarter

If your model mixes two groups up, add more examples of both, especially the kinds it got wrong. Then press Train again. This loop, **test, find mistakes, add better data, retrain**, is exactly what AI engineers do every day.

The same idea, much bigger

Real image AIs work like yours, but learn from millions of photos with much bigger models. India's postal service has used handwriting recognition to read PIN codes on letters, and apps like Google Lens recognise plants and objects from a photo.

What you learned

- An image classifier sorts pictures into groups you chose.
- You trained one by drawing labelled examples and pressing Train.
- Confidence shows how sure the model is; low confidence means it needs better examples.
- Test, find mistakes, add better data, retrain: that is how every AI improves.

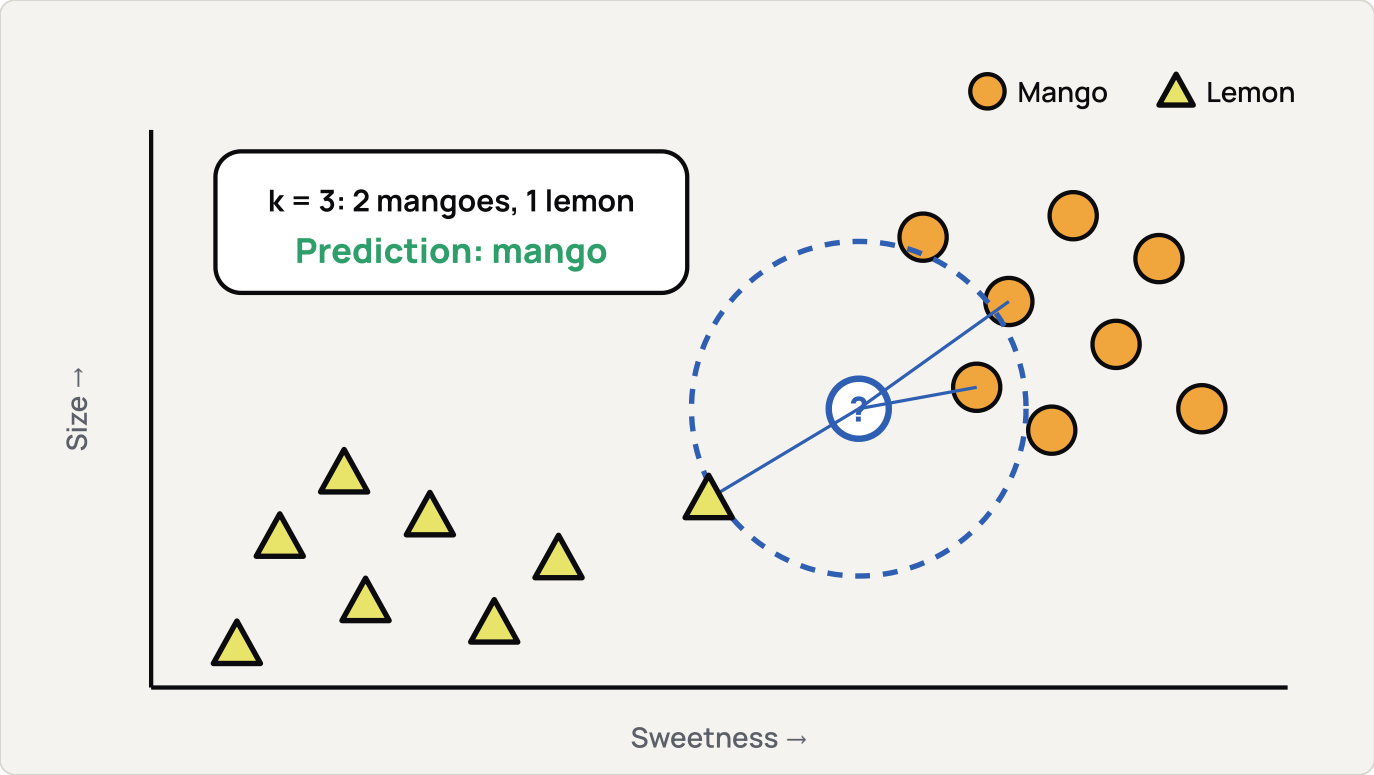
Next: Next, open up your model and see exactly how it makes its decisions.

How your AI decides: nearest neighbours

How did your model decide? It used one of the simplest and oldest ideas in AI: find the examples that look most like the new one, and go with the majority. It is called k-nearest neighbours.

You will:

- Explain how k-nearest neighbours makes a prediction
- See how changing k changes the answer
- Explain why examples near the border are hard



The new point (?) looks at its 3 nearest neighbours. Two are mangoes, one is a lemon, so the vote says mango.

Ask the neighbours

Imagine every example as a dot on a map. Fruits that are similar sit close together: sweet, big fruits in one corner, sour, small fruits in another.

When a new fruit arrives, the model:

1. Measures how far it is from every example.
2. Picks the **k** closest ones (k is just a number you choose, like 3).
3. Lets them **vote**. The group with the most votes wins.

Your drawing model did the same, except each drawing was a dot in a space with 256 directions, one for every pixel. Drawings with similar pixels sit close together.

▶ Online lab: drop points on a chart and watch the nearest neighbours vote.

New words

k-nearest neighbours (k-NN): predict by letting the k most similar examples vote.

Distance: how different two examples are. Small distance means very similar.

Decision boundary: the border where the prediction flips from one group to another.

What you learned

- k-nearest neighbours predicts by letting the k most similar examples vote.
- Similar examples sit close together; distance measures how different they are.
- Changing k can change the answer; an odd k avoids ties.
- Mistakes happen most near the border between groups.

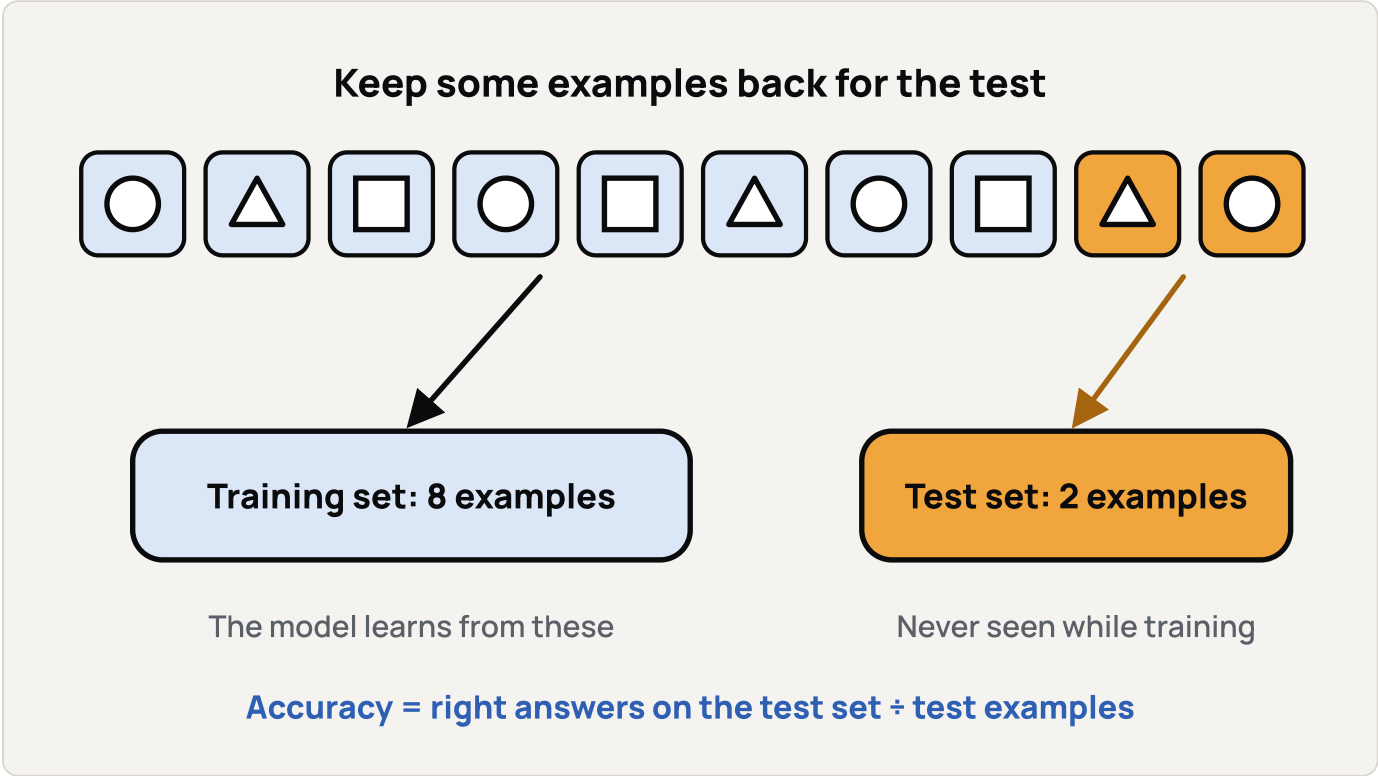
Next: Next, find out how to test a model properly, so you know how good it really is.

Testing: is your AI any good?

A model that only works on the examples it was trained on is like a student who memorised last year's answers. In this lesson you will train a new model, with the camera or by drawing, then test it fairly on examples it has never seen.

You will:

- Explain why you test on new examples
- Work out a model's accuracy
- Read a confusion matrix to find which groups get mixed up
- Improve a model and measure the difference



Keep some examples back. Train on the rest, then test on the ones the model has never seen.

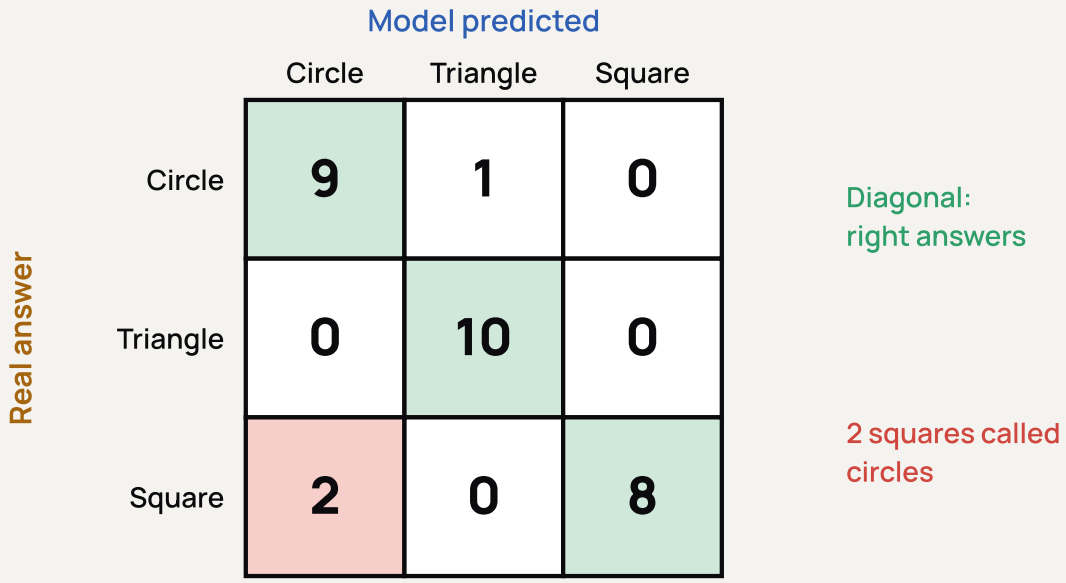
A fair test

To find out how good a model really is, you test it on **new examples it has never seen**, just like an exam has new questions.

Accuracy is the share it gets right. If it gets 9 out of 10 test examples right, its accuracy is 90%.

Accuracy alone does not tell you *which* mistakes it makes. For that, use a **confusion matrix**: a table showing, for each real group, what the model predicted.

Confusion matrix



Rows are the real answer, columns are the prediction. The diagonal shows the right answers; everything else is a mix-up.

▶ **Online lab: train your own model by drawing or with the camera, then test it.**

Your camera, your privacy

Camera pictures are shrunk to a tiny 12 × 9 grid and used only inside this page. They are never uploaded and are forgotten when you leave. If you prefer not to use a camera, switch to drawing: pick two shapes or symbols instead.

What you learned

- Test on new examples the model has never seen, like a fair exam.
- Accuracy is the percentage of test examples it gets right.
- A confusion matrix shows which groups get mixed up.
- Fix mistakes with more varied data, then retrain and retest.

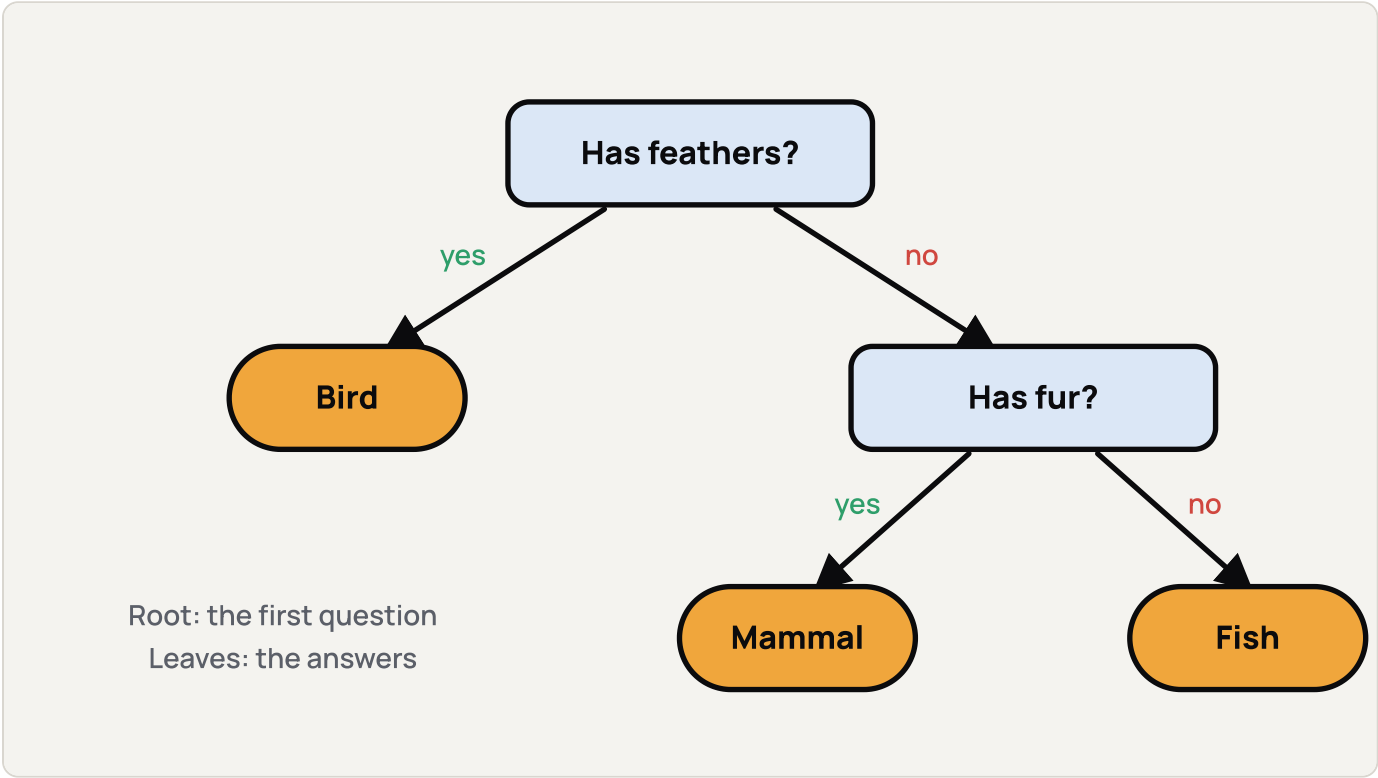
Next: Next, meet a different kind of model: a decision tree that learns by asking questions.

Decision trees: AI that asks questions

Remember the game where you guess an animal by asking yes-or-no questions? A decision tree is an AI that plays that game. It is one of the easiest models to understand, because you can read every decision it makes.

You will:

- Explain how a decision tree makes a prediction
- Build a tree that sorts animals by asking good questions
- Explain why a tree that is perfect on training data can still fail on new data



Start at the top. Answer each question and follow the branch until you reach an answer.

Twenty questions, for computers

A **decision tree** starts with one question at the top (the **root**). Each answer leads down a **branch** to another question, until you reach a **leaf**: the prediction.

The skill is choosing good questions. A good question splits the examples into groups that are as pure as possible. “Does it have feathers?” is brilliant for finding birds. “Is it brown?” is not, because birds, fish and mammals can all be brown.

Computers build trees by trying every possible question and keeping the one that splits the data best. You will do it by hand.

▶ Online lab: build a decision tree that sorts animals, then test it on unusual ones.

Trees you can trust

Because you can read every step, decision trees are used where people need to understand *why*: for example, helping decide which patients a nurse should see first, or whether a loan application needs a closer look. A person can check each question and spot unfair ones.

What you learned

- A decision tree predicts by asking a chain of yes-or-no questions.
- Good questions split the examples into pure groups.
- You can read every step, so people can check a tree's decisions.
- A tree perfect on training data can still fail on unusual new examples, like whales or penguins.

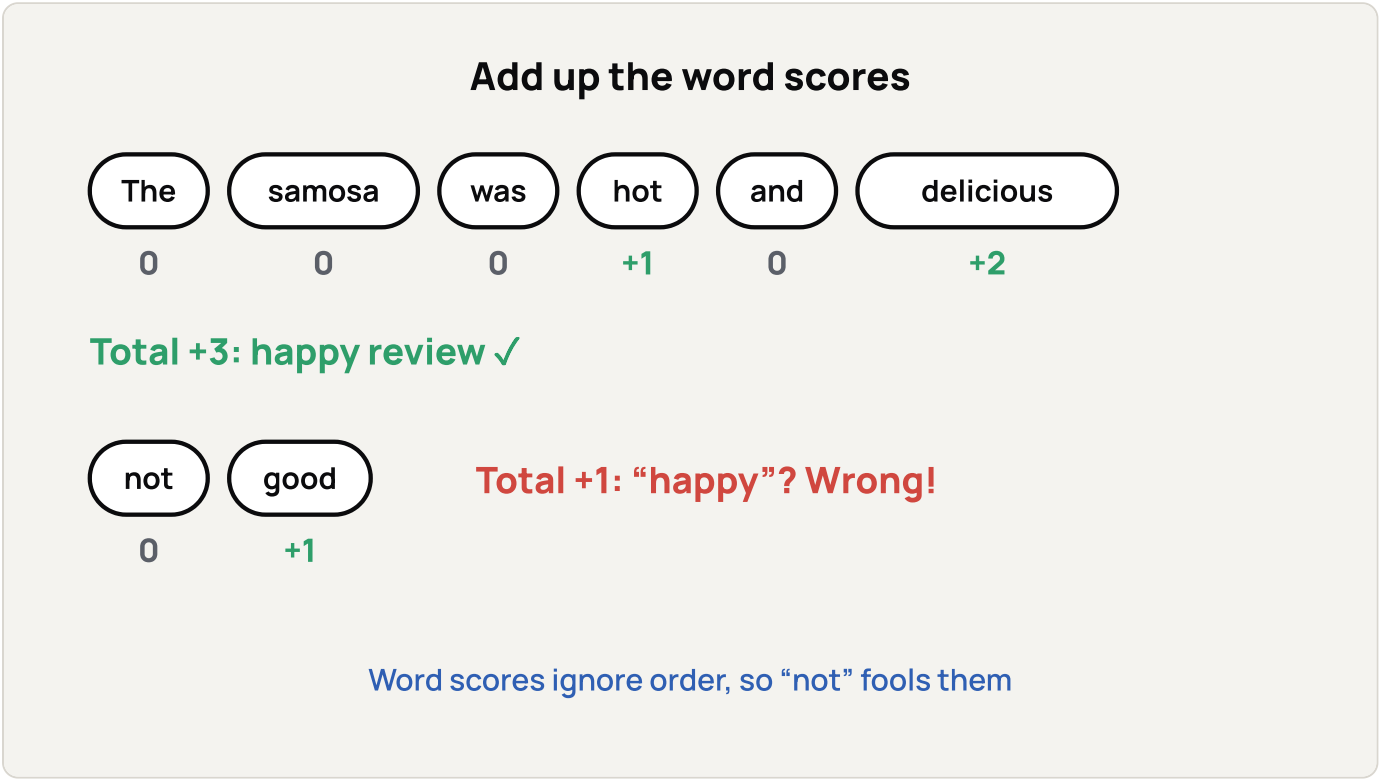
Next: Next, teach an AI to read: can it tell whether a message is happy or upset?

Teaching AI to read: words and feelings

Companies read thousands of reviews and messages every day to find out how people feel. That job is called sentiment analysis. In this lesson you will build a simple one, find where it breaks, and fix it.

You will:

- Explain how a computer can turn words into numbers
- Build a word-score model that sorts reviews into happy and unhappy
- Explain why words like “not” make language hard for AI



Give each word a score, add them up: above zero is happy, below zero is unhappy.

Words become numbers

Just as pictures become pixel numbers, words must become numbers for a computer. One simple way: give every word a **score**. “Delicious” might be +2, “cold” −1, “awful” −2, “the” 0.

To judge a review, add up the scores of its words. Positive total: happy. Negative total: unhappy. This is called **sentiment analysis**.

Real AI learns these scores from millions of reviews instead of a person typing them in, but the idea is the same.

▶ **Online lab: give words scores and teach the AI to read canteen reviews.**

Language is tricky

“Not bad at all” is a compliment, but adding up “not” and “bad” makes it sound negative. Sarcasm (“Oh great, cold samosas again”) is even harder. Modern AI looks at whole sentences and word order, not single words, which is one reason it needs so much data.

What you learned

- Words must become numbers for an AI; giving each word a score is one simple way.
- Adding word scores gives a sentiment: happy or unhappy.
- Words like “not”, and sarcasm, break simple word counting.
- Real language AI looks at whole sentences and learns from huge amounts of text.

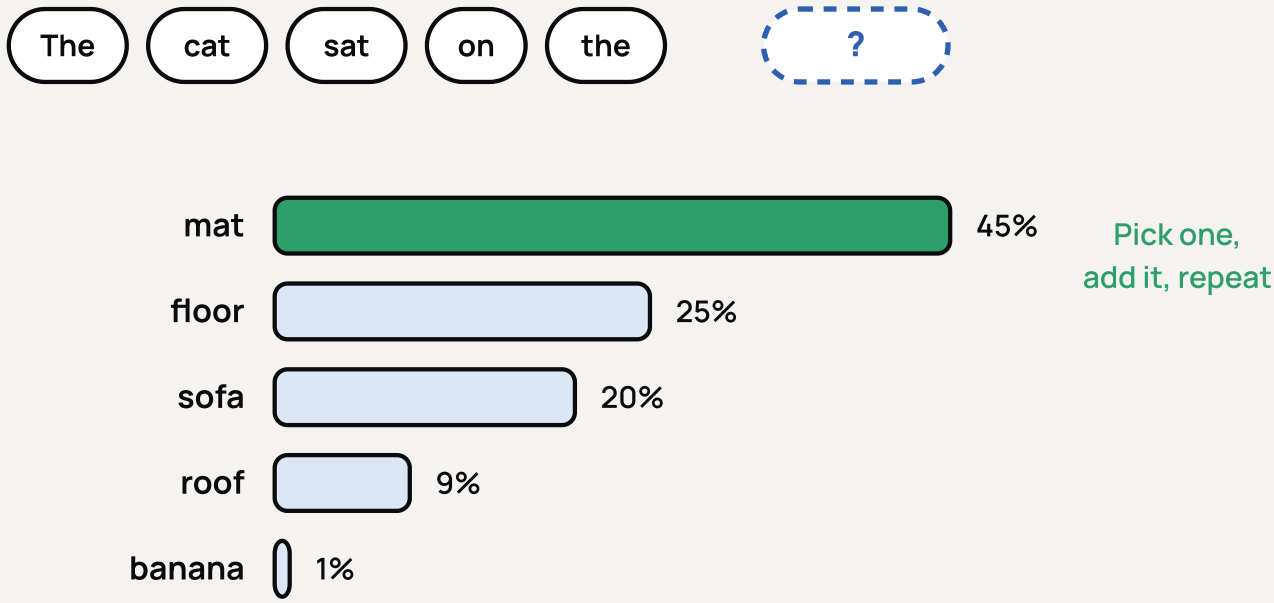
Next: Next, find out how chatbots write whole sentences: by predicting the next word, again and again.

How chatbots write: predicting the next word

Chatbots seem to think, but at their heart they do one thing over and over: predict the next word. In this lesson you will train a tiny language model on a short story and watch it write.

You will:

- Explain how a language model writes by predicting the next word
- Train a tiny language model and generate sentences with it
- Explain why chatbots can sound confident and still be wrong



After "The cat sat on the", the model gives every possible next word a chance, then picks one.

One word at a time

A **language model** has read a huge amount of text. From it, it learned which words tend to follow which. Give it "The cat sat on the" and it knows "mat", "floor" and "sofa" are likely, and "banana" is not.

To write an answer, it predicts one word, adds it, then predicts the next, and the next. Big chatbots learned from billions of sentences, so their predictions are very good. Your model will learn from one short story, so it will be much simpler, but it works the same way.

▶ **Online lab: a tiny language model suggests the next word, with its chances.**

Sounding right is not being right

A language model picks words that *sound* likely, not words it has checked are true. That is why chatbots sometimes invent facts, books or quotes that do not exist. People call this **hallucination**. Always check important facts somewhere you trust.

What you learned

- A language model writes by predicting the next word, again and again.
- It learns which words follow which from the text it was trained on.
- More training text gives better, more natural predictions.
- Likely-sounding is not the same as true, so check important facts.

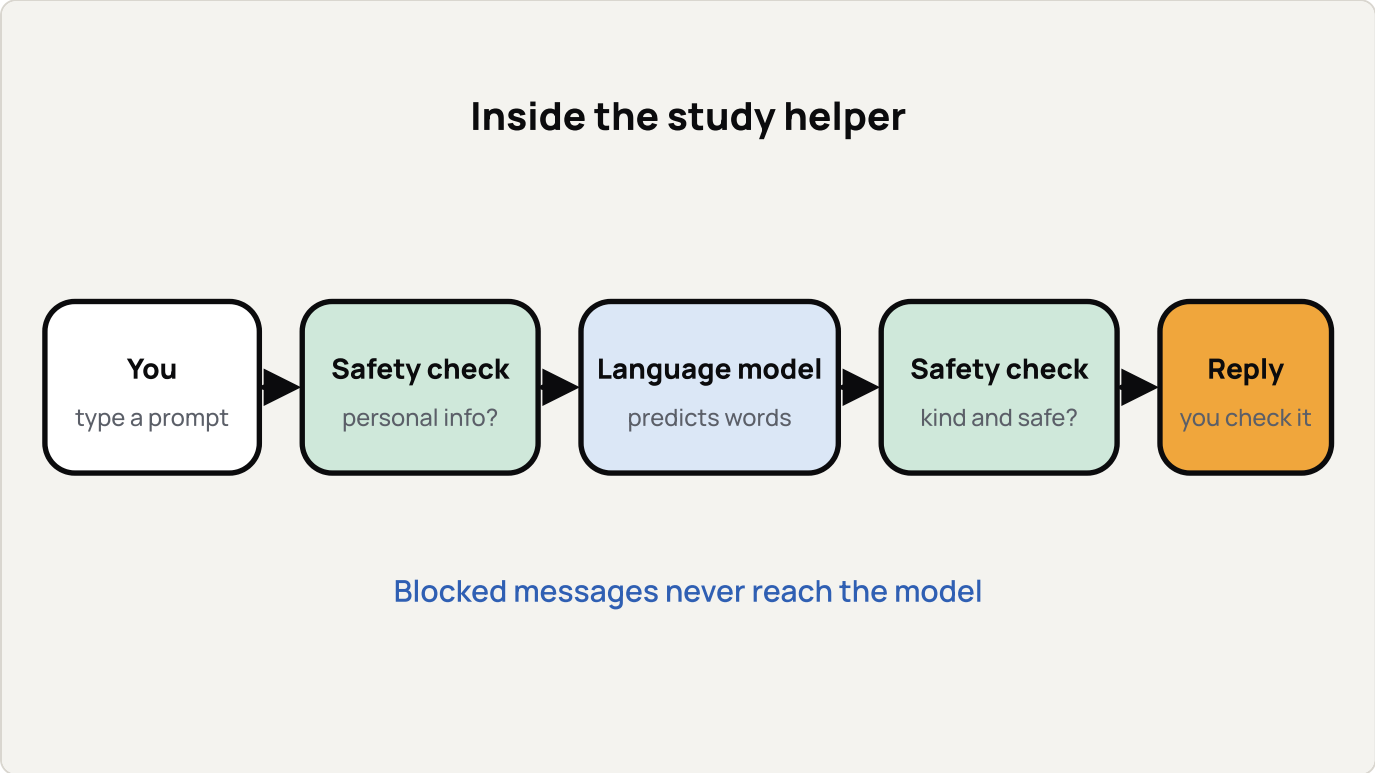
Next: Next, talk to a real AI chatbot, safely, and learn to ask it good questions.

Talk to a real AI, safely

Now you know how chatbots work, it is time to use one. You will talk to a real AI model through a safe study helper, learn to write clear prompts, and practise checking what it tells you.

You will:

- Write clear, specific prompts
- Use an AI to explain an idea, then check its answer
- Follow the Safe AI rules while chatting



Your message is checked for safety, the model predicts a reply, and the reply is checked again before you see it.

Prompts: how you ask matters

The message you give an AI is called a **prompt**. Clear prompts get better answers.

- **Vague:** "Tell me about space."
- **Clear:** "Explain in 3 short points why Mars looks red, for a Class 7 student."

A good prompt says **what** you want, **how long**, and **who it is for**. If the answer is not right, ask again more clearly, or ask it to explain more simply.

Before you chat

This helper is for learning. It will not answer questions that are unsafe, and messages that contain personal details are blocked before they reach the AI. Your teacher can see how much the class uses it. **Never type your name, address, phone number, school or passwords.** If anything worries you, stop and tell an adult.

▶ **Online lab: chat with the safe study helper and complete three prompt missions.**

What you learned

- A prompt is the message you give an AI; clear prompts get better answers.
- Say what you want, how long, and who it is for.
- Keep personal details out of every prompt.
- Check important facts from AI with a trusted source.

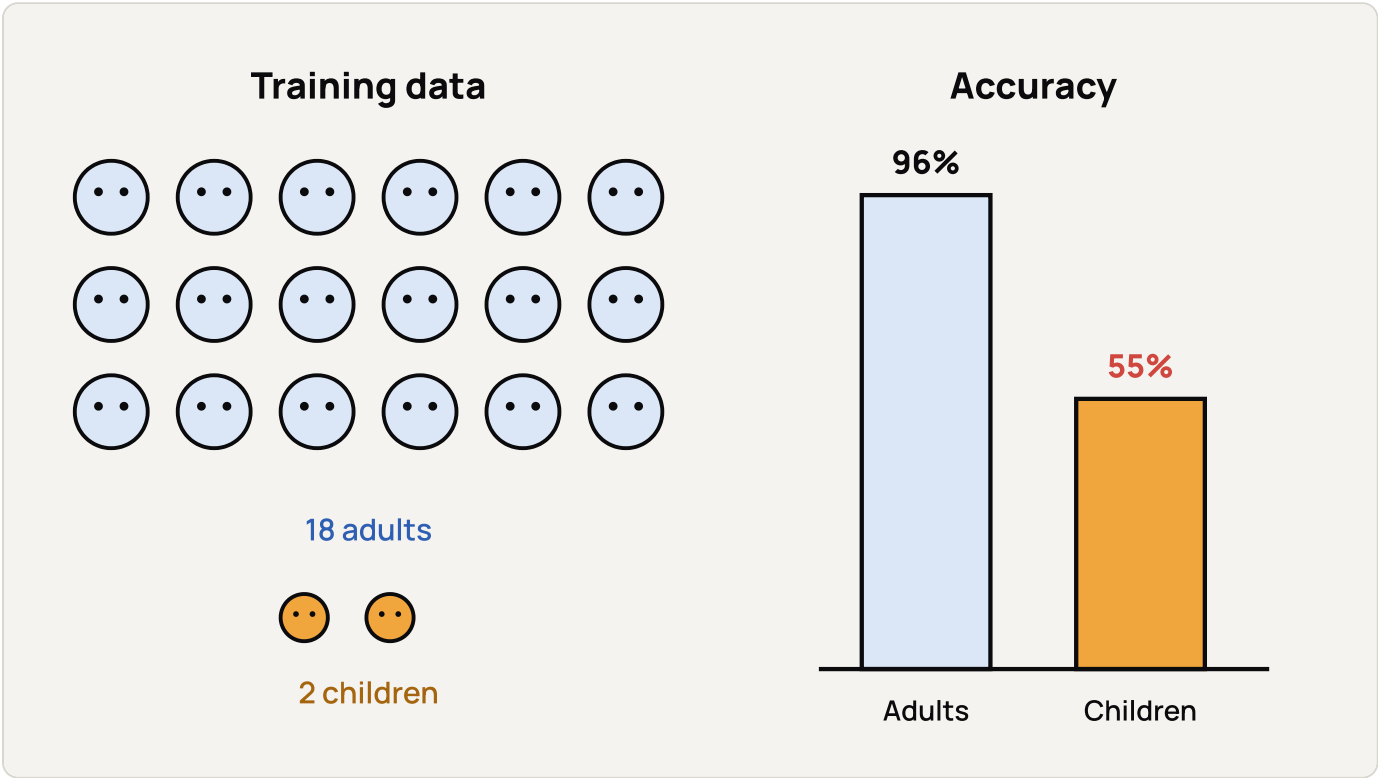
Next: Next, find out how AI can be unfair, and what you can do about it.

Fair AI: when data is unfair

An AI learns exactly what its data teaches it, including the unfair parts. In this lesson you will see a model become unfair because of its data, and then fix it.

You will:

- Explain what bias in AI means
- Show how unbalanced data makes a model unfair
- Suggest ways to make data, and AI, fairer



Lots of examples of one group and very few of another: the model learns the big group well and gets the small group wrong.

Unfair data, unfair AI

Bias means an AI treats some people or things unfairly, usually because of its data.

Imagine a face-unlock AI trained mostly on photos of adults. It may work well for adults and badly for children. Nobody meant it to be unfair: the data simply had too few children in it.

Bias can also come from history. If an AI learns from old records where some groups were treated unfairly, it can copy that unfairness into the future.

▶ **Online lab: drop points on a chart and watch the nearest neighbours vote.**

Real stories

Researchers have found face recognition systems that made far more mistakes on darker-skinned faces, because they were trained mostly on lighter-skinned faces. Speech assistants have struggled with Indian accents when trained mostly on American and British voices. Companies improved them by collecting more varied data.

What you learned

- Bias means an AI treats some people or things unfairly, usually because of its data.
- Unbalanced data makes a model good at the big group and bad at the small one.
- Check accuracy for each group, not just overall.
- More balanced, more varied data is the main fix.

Next: Next, learn to use AI responsibly: privacy, fake images and being honest about AI help.

Responsible AI: privacy, deepfakes and you

AI is a powerful tool, and like any tool it can be used well or badly. In this lesson you will practise the choices that keep you and others safe: protecting privacy, spotting fakes and being honest about AI help.

You will:

- Decide what information is safe to share with AI
- Spot signs of AI-made fake images, videos and voices
- Use AI honestly in schoolwork
- Make your own Responsible AI promise

Three big questions

Privacy: whose data is it? AI apps may keep what you type or upload. Share as little as possible, and never share other people's photos or details without asking them.

Truth: is it real? AI can make **deepfakes**: fake photos, videos and voices of real people. Before you believe or share something surprising, stop and check where it came from.

Honesty: who did the work? Using AI to brainstorm or explain is fine when your teacher allows it. Copying AI answers and calling them your own is not. Always say when and how you used AI.

Spotting a fake: STOP

Source: who posted it first? A trusted news site, or an unknown account?

Too surprising? Shocking content is made to be shared fast. Slow down.

Odd details: strange hands, blurry edges, words that do not match lip movements, a voice that sounds flat.

Proof: can you find the same story on two trusted sites? If not, do not share it.

My Responsible AI promise

- [ ] I keep my personal details, and other people's, private
- [ ] I check important facts from AI with a trusted source
- [ ] I stop and check before I share surprising pictures or videos
- [ ] I never use AI to trick, tease or hurt anyone
- [ ] I say when and how I used AI in my work
- [ ] I tell an adult if AI shows me something worrying

What you learned

- Share as little personal information with AI as possible, and never other people's.
- AI can make convincing fakes: check the source before you believe or share.
- Use AI honestly: for ideas and explanations, and say when you used it.
- Never use AI to hurt or trick anyone.

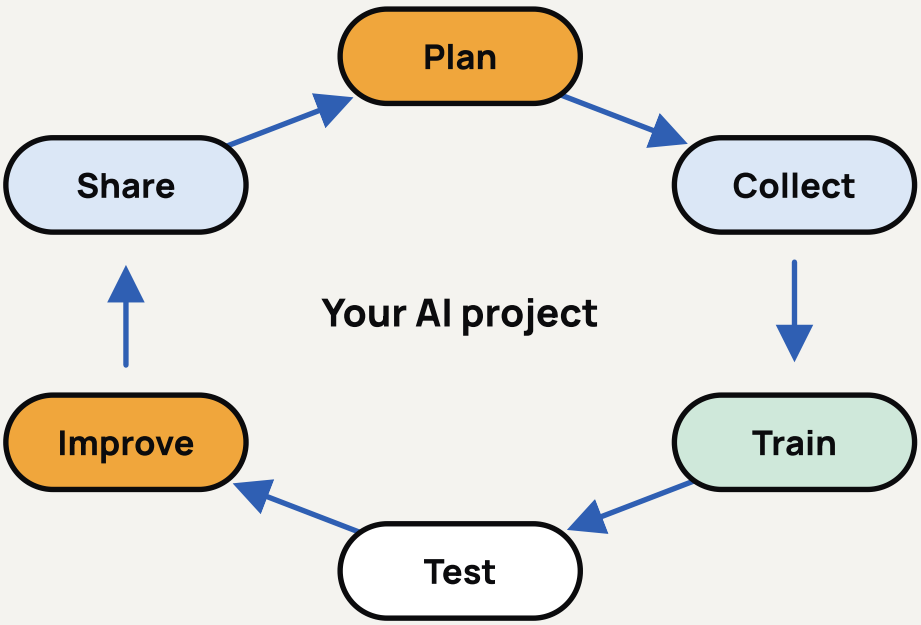
Next: Next, the final project: build, test and present your own AI.

Final project: build and present your AI

This is your moment. Use everything you have learned to build an AI that solves a problem you choose. Plan it, collect your data, train, test, improve, then hand it in for your teacher to review.

You will:

- Build a classifier for a problem you choose
- Test it fairly and report accuracy honestly
- Make and measure one improvement
- Explain who your AI might be unfair to, and how you would check



Every AI project goes round this loop: plan, collect, train, test, improve, share.

Project steps

- [ ] Choose a problem and two to four groups
- [ ] Collect varied examples for every group
- [ ] Train your model
- [ ] Test it on new examples and note the accuracy
- [ ] Read the confusion matrix and find the biggest mix-up
- [ ] Improve the data, retrain and test again
- [ ] Think about fairness: who might it fail for?
- [ ] Write your report and hand it in

▶ Online lab: train your own model by drawing or with the camera, then test it.

What you learned

- You planned, built, tested and improved your own AI.
- You reported accuracy honestly and found its biggest mix-up.
- You thought about who your AI might fail for.

Next: Last step: the final assessment, your badge and your certificate.

Final assessment and celebration

Twelve questions on everything you have learned. You need 80% to pass, and you can try again as many times as you like. Your best score counts.

You will:

- Show what you know about data, models, testing, language AI and fairness
- Earn the AI Builder badge and your certificate

Where next?

Try **Robotics 101** to give your AI a body, or explore tools like Teachable Machine and Scratch's machine-learning extensions with your teacher. Keep asking: where did the data come from, and is it fair?

What you learned

- You trained, tested and improved your own AI, and learned how chatbots write.
- Good AI needs good, fair data, honest testing and responsible use.

Next: You have finished AI for Young Builders. Download your certificate from your learning page.